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Generalization of Newman normalization to weighted two-mode networks

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<https://github.com/bavla/Nets>

Network = Graph + Data

A *network* $\mathcal{N} = (\mathcal{V}, \mathcal{L}, \mathcal{P}, \mathcal{W})$ consists of:

- a *graph* $\mathcal{G} = (\mathcal{V}, \mathcal{L})$, where $\mathcal{V}$ is the set of nodes, $\mathcal{A}$ is the set of arcs (directed links), $\mathcal{E}$ is the set of edges (undirected links), and $\mathcal{L} = \mathcal{E} \cup \mathcal{A}$ is the set of links. $n = |\mathcal{V}|$, $m = |\mathcal{L}|$
- $\mathcal{P}$ *node value functions* / properties: $p: \mathcal{V} \rightarrow A$
- $\mathcal{W}$ *link value functions* / weights: $w: \mathcal{L} \rightarrow B$

In a *two-mode* network $\mathcal{N} = ((\mathcal{U}, \mathcal{V}), \mathcal{L}, \mathcal{P}, \mathcal{W})$ the set of nodes consists of two disjoint sets of nodes $\mathcal{U}$ and $\mathcal{V}$, and all the links from $\mathcal{L}$ have one end node in $\mathcal{U}$ and the other in $\mathcal{V}$.

$$\text{wod}(v) = \sum_{e \in \text{ostar}(v)} w(e) \quad \text{and} \quad \text{wid}(v) = \sum_{e \in \text{istar}(v)} w(e)$$

where $\text{ostar}(v)$ is the out-star – links leading out of node v , etc.

Weighted network matrix

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$$\text{wto}(v, u) = \sum_{e \in \text{ostar}(v) \cap \text{istar}(u)} w(e)$$

A *weighted two mode network* $\mathcal{N} = ((\mathcal{U}, \mathcal{V}), \mathcal{L}, w)$ can be represented with the corresponding *matrix* $\mathbf{UV} = [uv[u, v]]_{u \in \mathcal{U}, v \in \mathcal{V}}$ defined as

$$uv[u, v] = \begin{cases} \text{wto}(u, v) & (u, v) \in \mathcal{L} \\ \square & \text{otherwise} \end{cases}$$

We extend the set of weight values with a new zero element $\square$ (no link) with rules $\square + a = a$ and $\square \cdot a = \square$.

In the case $\mathcal{U} = \mathcal{V}$ we obtain a square matrix of an ordinary network.

A network $\mathcal{N}$ is *binary* iff the weight w is constant $w = 1$.

The *transpose* $\mathbf{UV}^T$ of matrix $\mathbf{UV}$ is a matrix on $\mathcal{U} \times \mathcal{V}$ with entries $uv^T[v, u] = uv[u, v]$. $\mathbf{VU} = \mathbf{UV}^T$

Reversed network $\tilde{\mathcal{N}} = ((\mathcal{V}, \mathcal{U}), \tilde{\mathcal{L}}, vu)$, $\tilde{\mathcal{L}} = \{(v, u) : (u, v) \in \mathcal{L}\}$, $vu(v, u) = uv(u, v)$ for $(u, v) \in \mathcal{L}$.

$$\text{wod}_{VU}(v) = \text{wid}_{UV}(v), \dots$$

In the following, we will often identify networks by their matrices.

The *total weight* of links in the network $\mathcal{N} = (\mathcal{V}, \mathcal{L}, w)$ with matrix $\mathbf{N} = [n[u, v]]$

$$T(\mathbf{N}) = \sum_{(u,v) \in \mathcal{L}} w(u, v) = \sum_{u,v} n[u, v] = \sum_u \text{wod}_N(u) = \sum_v \text{wid}_N(v)$$



Some network transformations

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$S(\mathbf{N})$ transforms a network $\mathbf{N}$ into a simple undirected network with link weights equal to the sum of corresponding arc weights.

$D_0(\mathbf{N})$ removes loops from a network $\mathbf{N}$.

Binarization $\text{bin}(\mathbf{N})$ sets the weight of each link to 1.



Multiplication of networks

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Given a pair of *compatible* networks $\mathcal{N}_A = ((\mathcal{I}, \mathcal{K}), \mathcal{A}_A, w_A)$ and $\mathcal{N}_B = ((\mathcal{K}, \mathcal{J}), \mathcal{A}_B, w_B)$ with corresponding matrices $\mathbf{A}_{\mathcal{I} \times \mathcal{K}}$ and $\mathbf{B}_{\mathcal{K} \times \mathcal{J}}$ we call a *product of networks* $\mathcal{N}_A$ and $\mathcal{N}_B$ a network $\mathcal{N}_C = ((\mathcal{I}, \mathcal{J}), \mathcal{A}_C, w_C)$, where $\mathcal{A}_C = \{(i, j) : i \in \mathcal{I}, j \in \mathcal{J}, c_{i,j} \neq 0\}$ and $w_C(i, j) = c_{i,j}$ for $(i, j) \in \mathcal{A}_C$. The product matrix $\mathbf{C} = [c_{i,j}]_{\mathcal{I} \times \mathcal{J}} = \mathbf{A} * \mathbf{B}$ is defined in the standard way

$$c_{i,j} = \sum_{k \in \mathcal{K}} a_{i,k} \cdot b_{k,j}$$

In the case when $\mathcal{I} = \mathcal{K} = \mathcal{J}$ we are dealing with ordinary one-mode networks (with square matrices).

Multiplication of networks

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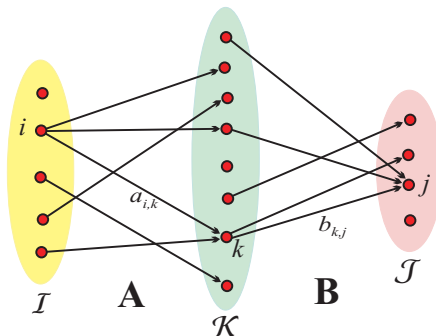
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$$c_{i,j} = \sum_{k \in oN_A(i) \cap iN_B(j)} a_{i,k} \cdot b_{k,j}$$

If both networks $\mathcal{N}_A$ and $\mathcal{N}_B$ are binary the value of $c_{i,j}$ counts the number of ways we can go from $i \in \mathcal{I}$ to $j \in \mathcal{J}$ passing through $\mathcal{K}$, $c_{i,j} = |oN_A(i) \cap iN_B(j)|$. oN are out-neighbors, etc.



Projections

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Two-mode matrices $\mathbf{UV}$ and $\mathbf{UV}^T$ are always compatible on both sides. The corresponding derived one-mode matrices are called *projections*

- *row* projection $\mathbf{VV} = \text{row}(\mathbf{UV}) = \mathbf{UV}^T \cdot \mathbf{UV}$
- *column* projection $\mathbf{UU} = \text{col}(\mathbf{UV}) = \mathbf{UV} \cdot \mathbf{UV}^T$

They are related $\text{row}(\mathbf{UV}) = \text{col}(\mathbf{UV}^T) = \text{col}(\mathbf{VU})$.

Therefore, it is enough to study the properties of one of them.

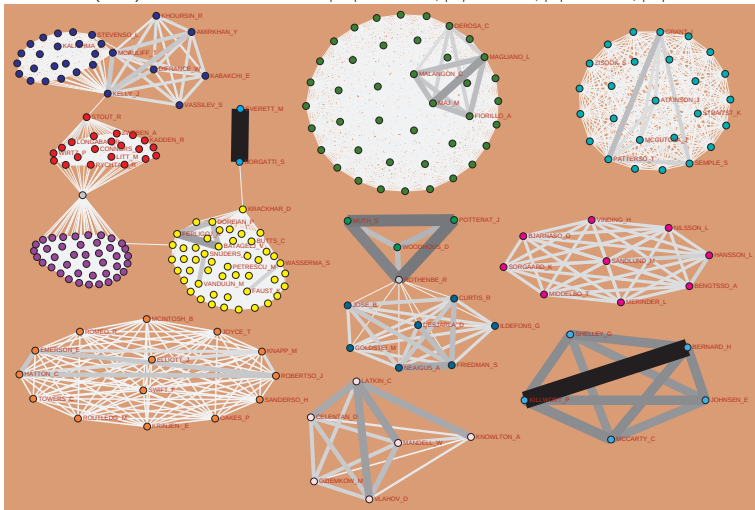
Let $\mathbf{Co} = \text{row}(\mathbf{UV})$ – the *co-occurrence* network. It is symmetric $\text{Co}[v, z] = \text{Co}[z, v]$.

p_5 -core at level 20 of **Co(SN5)**

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Network SN5 (2008): for "social network*" $|W| = 193376$, $|A| = 75930$, $|J| = 14651$, $|K| = 29267$





Fractional approach

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For vectors $x = [x_1, x_2, \dots, x_n]$ and $y = [y_1, y_2, \dots, y_m]$ their *outer product* $x \circ y$ is defined as a matrix

$$x \circ y = [x_i \cdot y_j]_{n \times m}$$

then we can express the product $\mathbf{C}$ of two compatible matrices $\mathbf{A}$ and $\mathbf{B}$ as the *outer product decomposition*

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} = \sum_k \mathbf{H}_k \quad \text{where} \quad \mathbf{H}_k = \mathbf{A}[:, k] \circ \mathbf{B}[k, :],$$

$\mathbf{A}[:, k]$ is the k -th column of matrix $\mathbf{A}$, and $\mathbf{B}[k, :]$ is the k -th row of matrix $\mathbf{B}$.

$$T(x \circ y) = S_x \cdot S_y \quad \text{where} \quad S_x = \sum_i x_i$$

In general, each term $\mathbf{H}_k$ in the outer product decomposition has a different value $T(\mathbf{H}_k) = \text{wid}_A(k) \cdot \text{wod}_B(k)$ leading to over-representation of nodes with large values.

$$T(\mathbf{C}) = T\left(\sum_k \mathbf{H}_k\right) = \sum_k T(\mathbf{H}_k) = \sum_k \text{wid}_A(k) \cdot \text{wod}_B(k)$$

For $\mathbf{Co}$ we have

$$T(\mathbf{Co}) = \sum_{u \in \mathcal{U}} T(\mathbf{H}_u) = \sum_{u \in \mathcal{U}} \text{wod}(u)^2$$

To make the contributions of all nodes equal, we can apply the *fractional* approach by normalizing the weights: setting $x' = x/S_x$, we get $S_{x'} = 1$ and therefore $T(x' \circ y') = 1$.

Stochastic, Markov: $\hat{\mathcal{U}} = \{u \in \mathcal{U} : \text{wod}_{UV}(u) > 0\}$

$$n(UV)[u, v] = \begin{cases} \frac{uv[u, v]}{\text{wod}_{UV}(u)} & u \in \hat{\mathcal{U}} \\ \square & u \notin \hat{\mathcal{U}} \end{cases} \quad \text{wod}_{n(UV)}(u) = \begin{cases} 1 & u \in \hat{\mathcal{U}} \\ 0 & u \notin \hat{\mathcal{U}} \end{cases}$$

$$T(n(\mathbf{UV})) = \sum_{u \in \mathcal{U}} \text{wod}_{n(UV)}(u)^2 = \sum_{u \in \hat{\mathcal{U}}} 1 = |\hat{\mathcal{U}}|$$

Strict, Newman: for binary networks, self excluded

$\text{wod}'_{UV}(u, v) = \text{wod}_{UV}(u) - 1, \quad \mathcal{U}' = \{u \in \mathcal{U} : \text{wod}'_{UV}(u) > 0\}$

$$s(UV)[u, v] = \begin{cases} \frac{uv[u, v]}{\text{wod}'_{UV}(u)} & u \in \mathcal{U}' \\ \square & u \notin \mathcal{U}' \end{cases} \quad \text{wod}_{s(UV)}(u) = \begin{cases} 1 & u \in \mathcal{U}' \\ 0 & u \notin \mathcal{U}' \end{cases}$$

$$T(s(\mathbf{UV})) = \sum_{u \in \mathcal{U}} \text{wod}_{s(UV)}(u)^2 = \sum_{u \in \mathcal{U}'} 1 = |\mathcal{U}'|$$



Normalizations

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Weight-preserving normalization

$$p(UV)[u, v] = \begin{cases} \frac{uv[u, v]}{\sqrt{\text{wod}_{UV}(u)}} & u \in \hat{U} \\ \square & u \notin \hat{U} \end{cases}$$

$$\text{wod}_{p(UV)}(u) = \begin{cases} \sqrt{\text{wod}_{UV}(u)} & u \in \hat{U} \\ 0 & u \notin \hat{U} \end{cases}$$

$$T(u) = \text{wod}_{p(UV)}(u)^2 = \text{wod}_{UV}(u)$$

Weighted projections

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Fractional projection $\mathbf{Cn} = S(n(\mathbf{UV})^T \cdot n(\mathbf{UV}))$
 $\mathbf{Cn}^T = \mathbf{Cn}$ and $T(\mathbf{Cn}) = |\hat{\mathcal{U}}|$.

Strict fractional projection $\mathbf{Cs} = S(D_0(s(\mathbf{UV})^T \cdot n(\mathbf{UV})))$
 $\mathbf{Cs}^T = \mathbf{Cs}$ and $T(\mathbf{Cs}) = |\mathcal{U}'|$.

Each active node has value 1 that is distributed over the arcs of the projection.

Weight-preserving projection $\mathbf{Cp} = S(p(\mathbf{UV})^T \cdot p(\mathbf{UV}))$
 $\mathbf{Cp}^T = \mathbf{Cp}$ and $T(\mathbf{Cp}) = T(\mathbf{UV})$.

The total weight in the original two-mode network is distributed over the arcs of the projection.

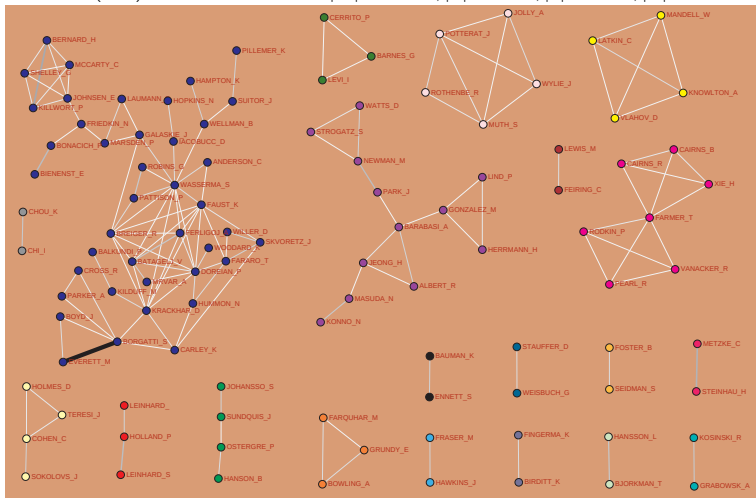


p_S -core at level 0.75 in **Cs(SN5)**

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Network SN5 (2008): for "social network*" $|W| = 193376$, $|A| = 75930$, $|J| = 14651$, $|K| = 29267$



Generalized Newman's normalization

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We assume the same form as in the standard Newman's normalization

$$s(UV)[u, v] = \begin{cases} \frac{uv[u, v]}{\text{wod}''_{UV}(u)} & u \in \mathcal{U}'' \\ \square & u \notin \mathcal{U}'' \end{cases}, \quad \mathcal{U}'' = \{u \in \mathcal{U} : \text{wod}''_{UV}(u) > 0\}$$

with an unknown function wod'' and expecting that in $\mathbf{Cs} = D_0(s(\mathbf{UV})^T \cdot n(\mathbf{UV}))$ for each node $u \in \mathcal{U}''$ we have

$$T_0(\mathbf{H}_u) = \sum_{v, z: v \neq z} H_u[v, z] = 1, \quad H_u[v, z] = \frac{uv[u, v]}{\text{wod}''(u)} \cdot \frac{uv[u, z]}{\text{wod}(u)}$$

We get

$$\begin{aligned} 1 &= T_0(\mathbf{H}_u) = \sum_{v, z: v \neq z} \frac{uv[u, v]}{\text{wod}''(u)} \cdot \frac{uv[u, z]}{\text{wod}(u)} = \\ &= \frac{1}{\text{wod}''(u) \cdot \text{wod}(u)} \sum_v (uv[u, v] \cdot \sum_{z: z \neq v} uv[u, z]) = \end{aligned}$$

... Generalized Newman's normalization

Since $\sum_{z: z \neq v} uv[u, z] = \text{wod}(u) - uv[u, v]$ we continue

$$= \frac{1}{\text{wod}''(u) \cdot \text{wod}(u)} (\text{wod}(u)^2 - \sum_v uv[u, v]^2)$$

Therefore

$$\text{wod}''(u) = \text{wod}(u) - \frac{\sum_v uv[u, v]^2}{\text{wod}(u)}$$

In the binary case $uv[u, v] \in \{0, 1\}$ we have $\text{wod}(u) = \text{od}(u)$, $\sum_v uv[u, v]^2 = \text{od}(u)$ and finally

$$\text{wod}''(u) = \text{od}(u) - \frac{\text{od}(u)}{\text{od}(u)} = \text{od}(u) - 1$$

which is the standard Newman's normalization.

Again we have $T_0(\mathbf{Cs}) = |\mathcal{U}''|$.

European protein consumption (1973)

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Manly, Bryan F. J., Navarro Alberto, Jorge A.: Multivariate statistical methods: a primer. 4e, CRC Press, 2017. page 198; Table 10.4 Sources of protein and percentages employed in different industry groups for countries in Europe.

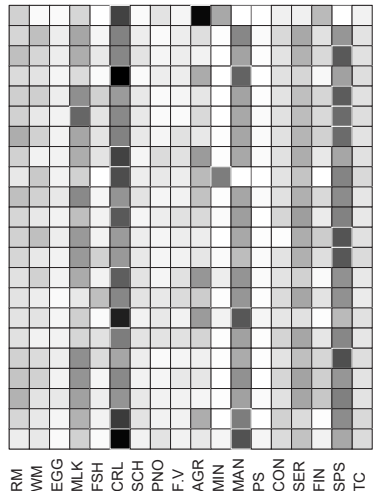
Estimates of the average protein consumption from different food sources for the inhabitants of 22 European countries (Weber, A. (1973). Agrarpolitik im Spannungsfeld der Internationalen Ernährungspolitik. Kiel: Institut für Agrarpolitik und Marktlehre.)

RM, red meat; WM, white meat; EGG, eggs; MLK, milk; FSH, fish; CLR, cereals; SCH, starchy foods; PNO, pulses, nuts, and oilseed; F&V, fruit and vegetables; AGR, agriculture, forestry, and fishing; MIN, mining and quarrying; MAN, manufacturing; PS, power and water supplies; CON, construction; SER, services; FIN, finance; SPS, social and personal services; TC, transport and communications.

[GitHub](#)

Pajek - shadow [0.00,57.00]

Albania
Austria
Belgium
Bulgaria
Denmark
Finland
France
Greece
Hungary
Ireland
Italy
Netherlands
Norway
Poland
Portugal
Romania
Spain
Sweden
Switzerland
UK
USSR
Yugoslavia



European protein consumption (1973)

Strongest 2-neighbors in Cs

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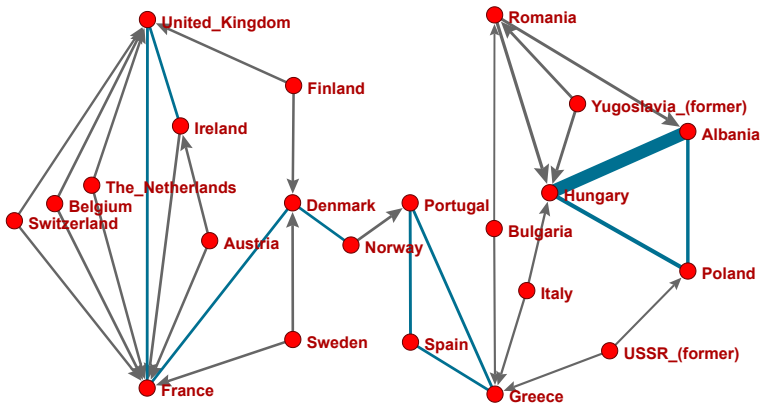
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Germany (East and West) is missing!!!



Conclusions

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- The obtained projections can be analyzed using standard network analysis methods [1, 5, 6].
- This is a traditional recipe for analyzing two-mode networks. Often, the weights are not considered in the analysis, and when they are, we must be careful about their meaning. In cases where the direct fractional approach is not meaningful, we can apply the approach proposed for normalized bibliographic coupling in [2, pp. 631–632].
- Traditional projections were defined for binary networks. We extended the definition to weighted networks.



Acknowledgments

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The computational work reported in this presentation was performed using R and the program for analysis and visualization of large networks [Pajek](#). The code and data are available at [GitHub/bavla](#).

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